

1. Lab Overview

In this lab, I practiced using **Group Policy** in a Windows Server 2019 Active Directory domain environment. The lab focused on creating and linking Group Policy Objects, applying desktop configuration settings, using `gpupdate`, and creating a mapped network drive for domain users.

This lab builds on the previous labs where I configured Active Directory, created users and groups, and configured shared folder permissions.

2. Business Scenario

This lab simulates a business or school environment where IT administrators need to centrally manage user desktop settings and provide users with access to shared network resources.

As the system administrator, I used Group Policy to configure desktop settings and map a shared network drive to the **SecurityPlus** folder. This allowed users to access a shared resource through the **S:** drive instead of manually typing the UNC path.

Your system administration lab includes creating a **Desktop Wallpaper** GPO, applying the policy with `gpupdate`, and creating a mapped network drive using a user configuration GPO.

3. Lab Objectives

The objectives of this lab were to:

- Open and use Group Policy Management Console.
- Create and link Group Policy Objects.
- Configure desktop settings through Group Policy.
- Apply Group Policy using `gpupdate`.
- Create a mapped network drive using Group Policy Preferences.
- Map the **SecurityPlus** shared folder to the **S:** drive.
- Configure permissions for Faculty and Students groups.
- Test mapped drive access using different domain users.

4. Lab Environment

Item	Details
Server VM	Windows Server 2019
Client VM	Windows 10

Domain	wara.edu
Shared Folder	SecurityPlus
Shared Folder UNC Path	\\WARA-SERVER\SecurityPlus
Mapped Drive	S: Drive
GPO Examples	Desktop Wallpaper, Clean Desktop, S: Drive
Main Tool	Group Policy Management Console
Commands Used	gpupdate

5. Tools Used

- Windows Server 2019
- Windows 10
- Active Directory Users and Computers
- Group Policy Management Console
- Group Policy Management Editor
- Group Policy Preferences
- File Explorer
- Command Prompt
- gpupdate
- Shared folder permissions
- NTFS permissions

6. Steps Performed

Step 1: Opened Group Policy Management Console

I opened **Group Policy Management Console** from Server Manager on the Windows Server 2019 VM.

This tool is used to create, edit, link, and manage Group Policy Objects in an Active Directory domain environment.

Step 2: Created and Linked a Desktop Wallpaper GPO

I created a new Group Policy Object called:

Desktop Wallpaper

This GPO was linked to the domain so that it could apply to domain users.

The lab used a shared image file named `bookcover.jpg` stored in the `SecurityPlus` shared folder. The wallpaper path was configured using the UNC path:

```
\\WARA-SERVER\SecurityPlus\bookcover.jpg
```

The lab directions show that the Desktop Wallpaper GPO was created, linked to the domain, edited in Group Policy Management Editor, and configured under User Configuration > Policies > Administrative Templates > Desktop.

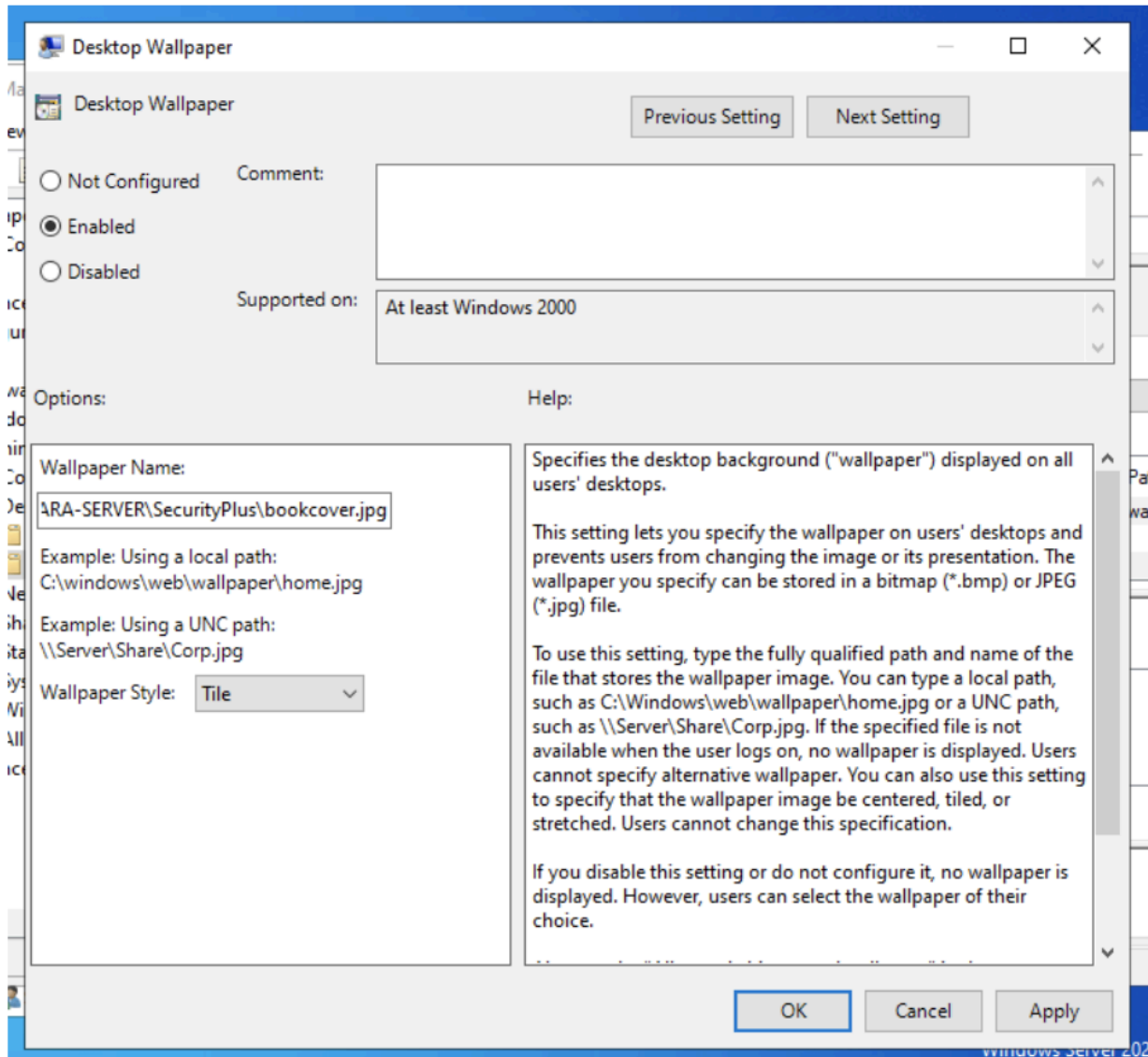


Figure 1: Desktop Wallpaper Group Policy configured with the shared UNC path.

Step 3: Applied the Group Policy Using gpupdate

After configuring the Desktop Wallpaper GPO, I applied the policy using Command Prompt on the Windows 10 VM.

Command used:

gpupdate

After signing out and signing back in, the wallpaper policy was applied to the user desktop.

The lab explains that after running `gpupdate`, users should sign out and sign back in to verify that the new wallpaper is visible on the desktops.

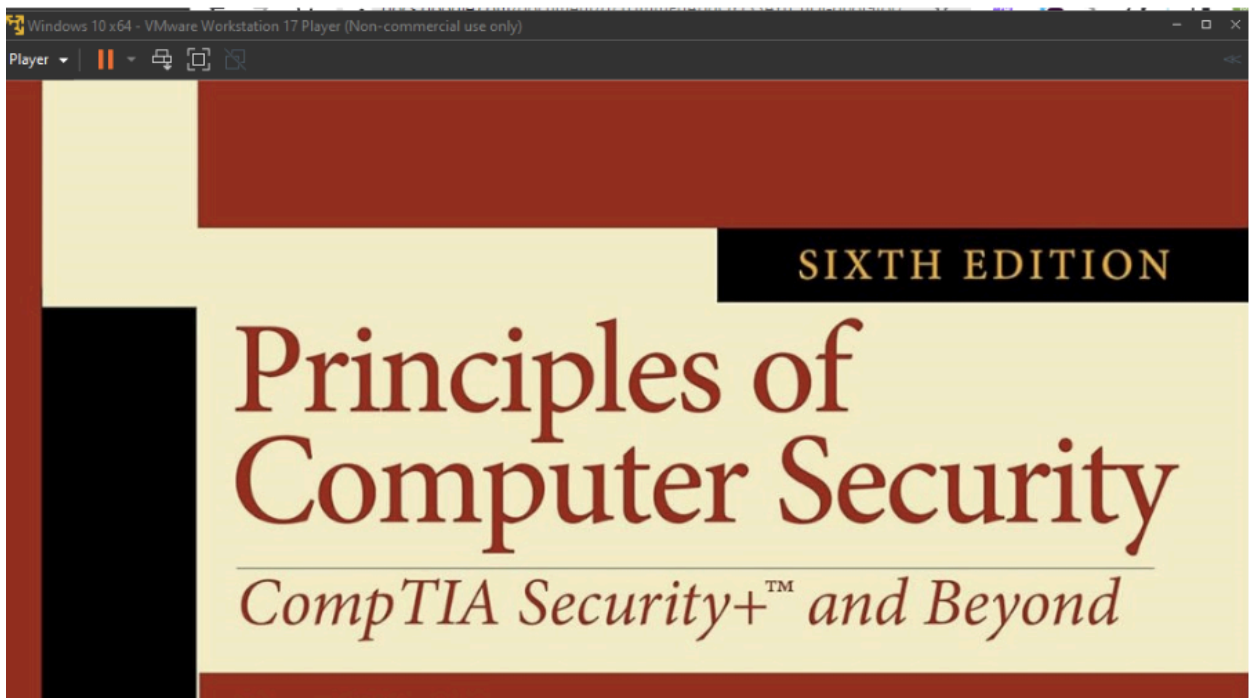
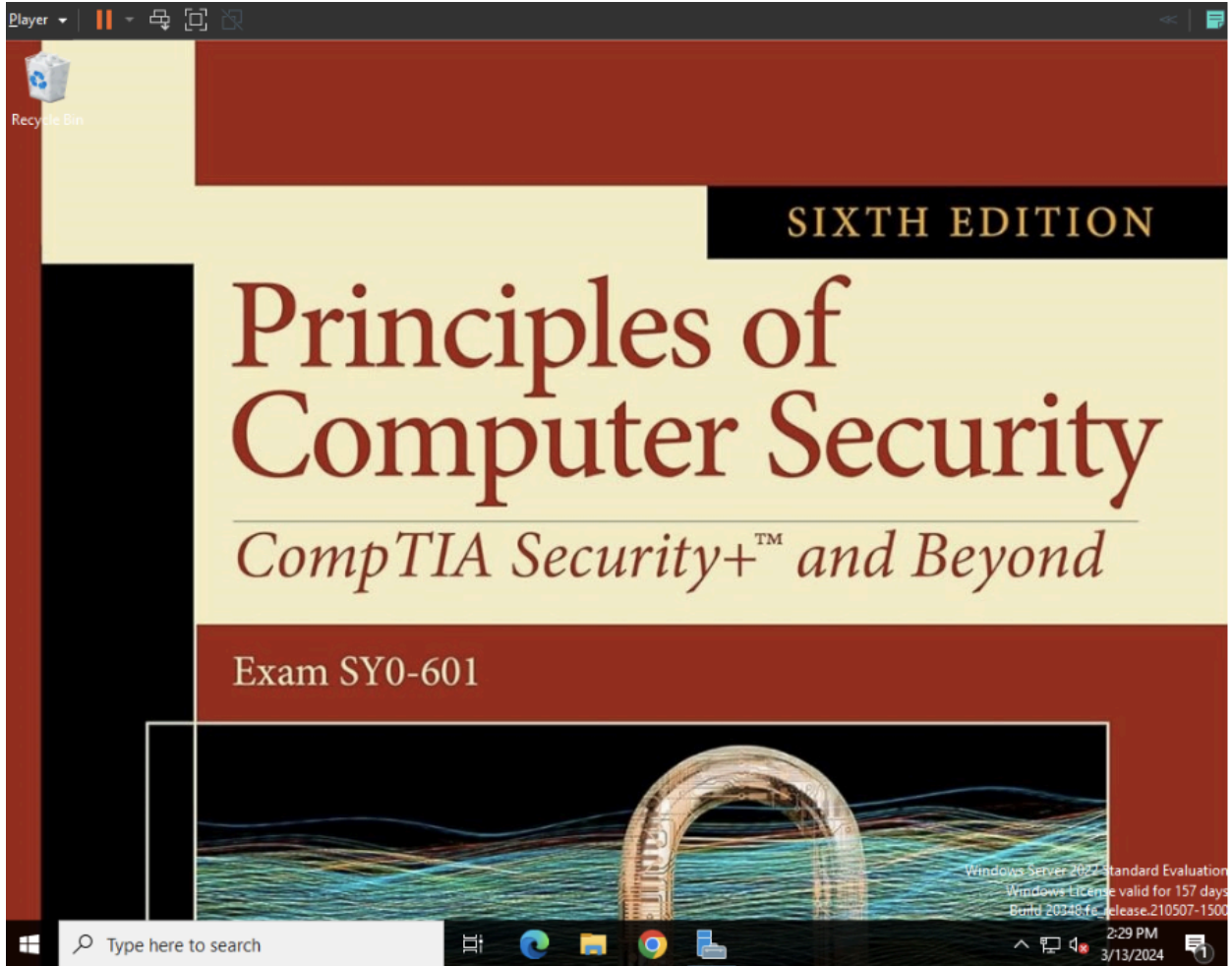


Figure 2: Group Policy updated and desktop configuration applied to the client machine.

Step 4: Configured a Clean Desktop GPO

I also configured a GPO called:

Clean Desktop

This GPO used the setting:

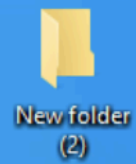
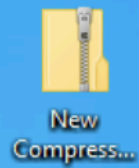
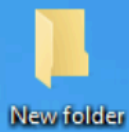
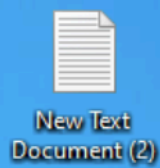
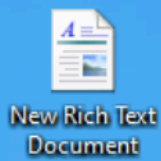
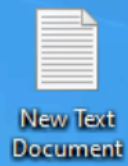
Hide and Disable All Items on the Desktop

The policy was configured under:

User Configuration > Policies > Administrative Templates > Desktop

Then it was linked to the Students OU so it would apply to student users.

The lab explains that the Clean Desktop GPO was created, configured, linked to the Students OU, and applied using `gpupdate` on the Windows 10 VM.



Hide and disable all items on the desktop

Hide and disable all items on the desktop

Previous Setting

Next Setting

☐ Not Configured

☒ Enabled

☐ Disabled

Comment:

Supported on:

At least Windows 2000

Options:

Help:

Removes icons, shortcuts, and other default and user-defined items from the desktop, including Briefcase, Recycle Bin, Computer, and Network Locations.

Removing icons and shortcuts does not prevent the user from using another method to start the programs or opening the items they represent.

Also, see "Items displayed in Places Bar" in User Configuration \Administrative Templates\Windows Components\Common Open File Dialog to remove the Desktop icon from the Places Bar. This will help prevent users from saving data to the Desktop.

OK

Cancel

Apply

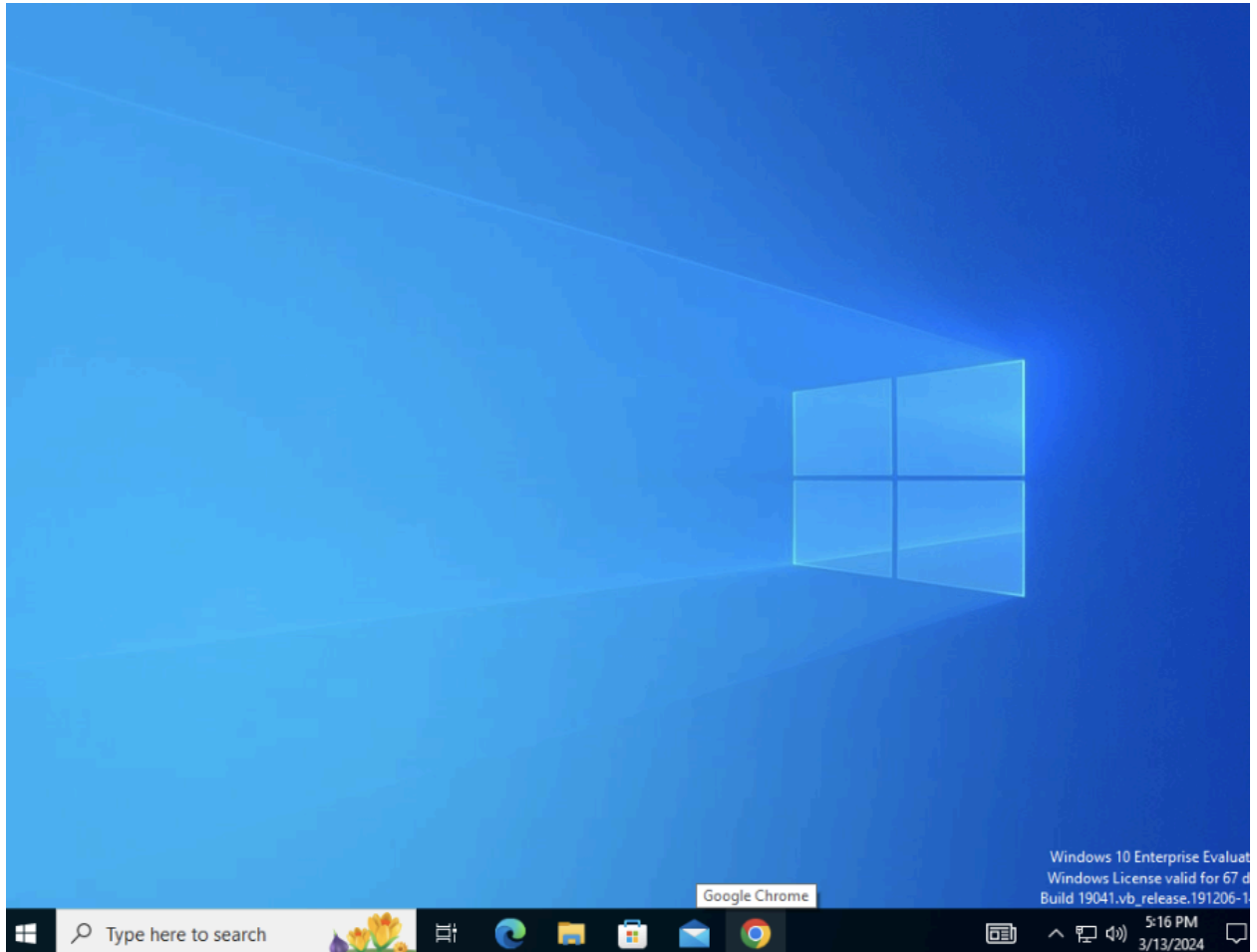


Figure 3: Clean Desktop GPO configured to hide and disable desktop items for student users.

Step 5: Created a Mapped Network Drive GPO

Next, I created a new Group Policy Object called:

S: Drive

This GPO was used to map the **SecurityPlus** shared folder to the **S:** drive for users.

The GPO was configured under:

User Configuration > Preferences > Windows Settings > Drive Maps

The mapped drive settings included:

Setting	Value
Action	Create or Update
Location	\\WARA-SERVER\SecurityPlus
Reconnect	Enabled
Label As	S: Drive
Drive Letter	S:

Your lab shows that the S: Drive GPO was created and configured through Group Policy Preferences > Drive Maps.

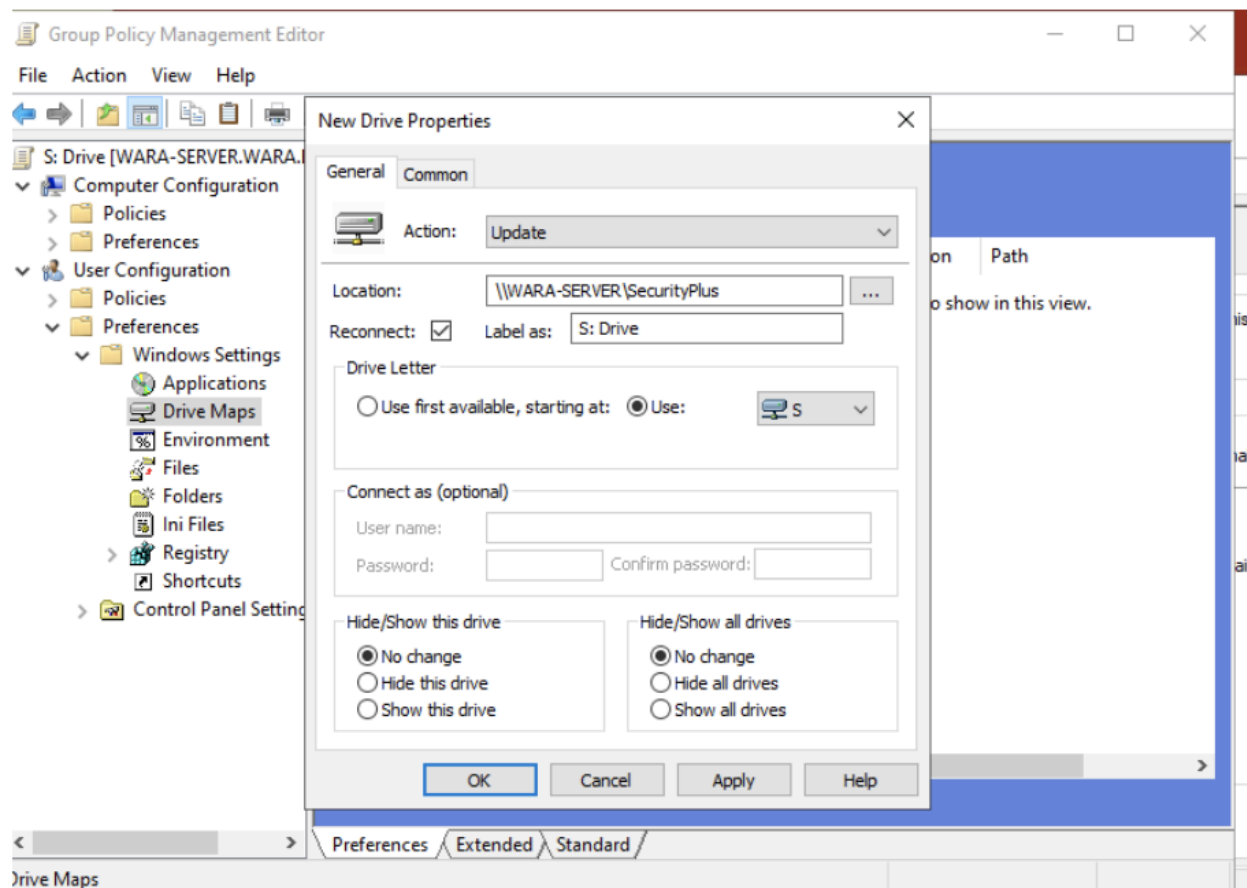


Figure 4: Group Policy Preferences configured to map the SecurityPlus share as the S: drive.

Step 6: Updated Group Policy and Configured Access Groups

After creating the mapped drive GPO, I ran `gpupdate` on both the Windows Server 2019 VM and Windows 10 VM.

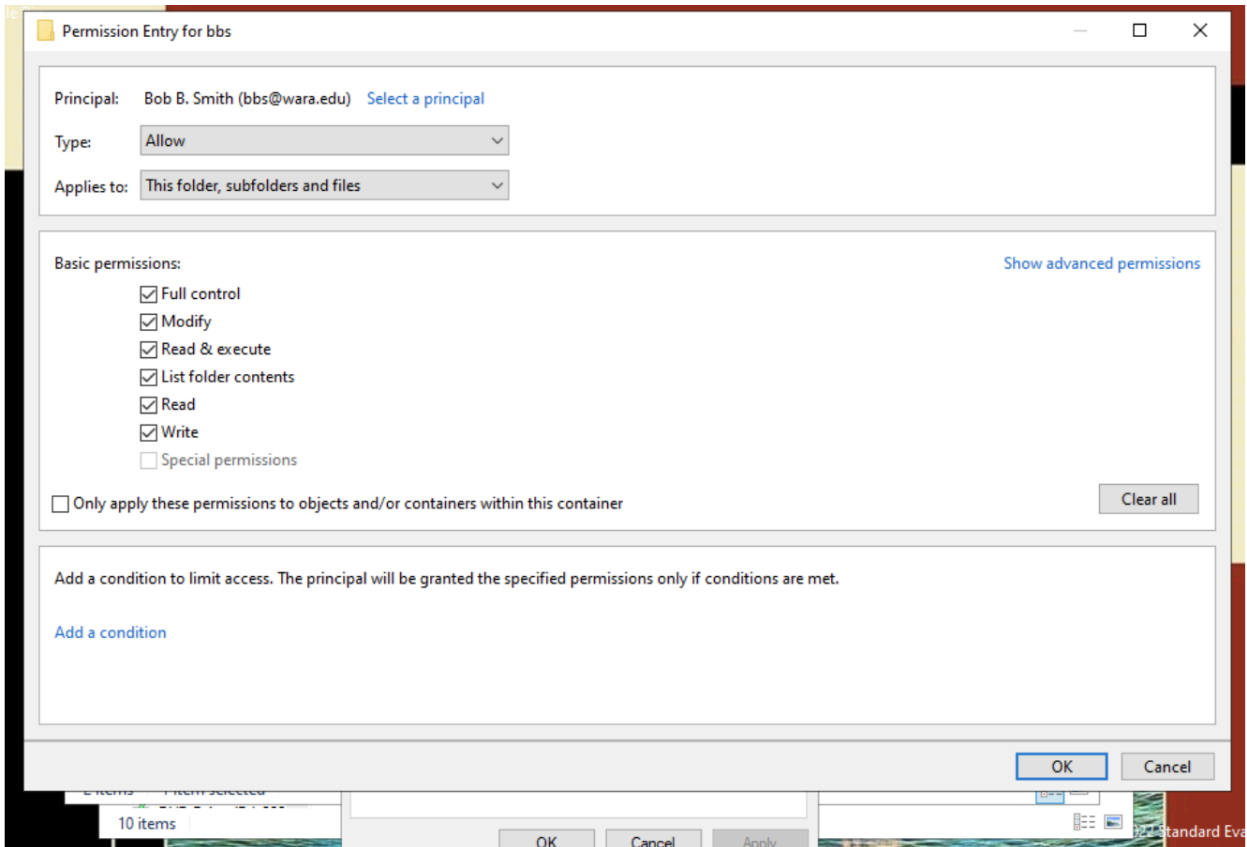
Then I created two domain local groups:

Faculty
Students

The permissions were configured like this:

Group	Share Permission	NTFS Permission
Faculty	Full Control	Full Control
Students	Read	Read

The lab directions state that Faculty should receive Full Control for both share and NTFS permissions, while Students should receive Read for both share and NTFS permissions.



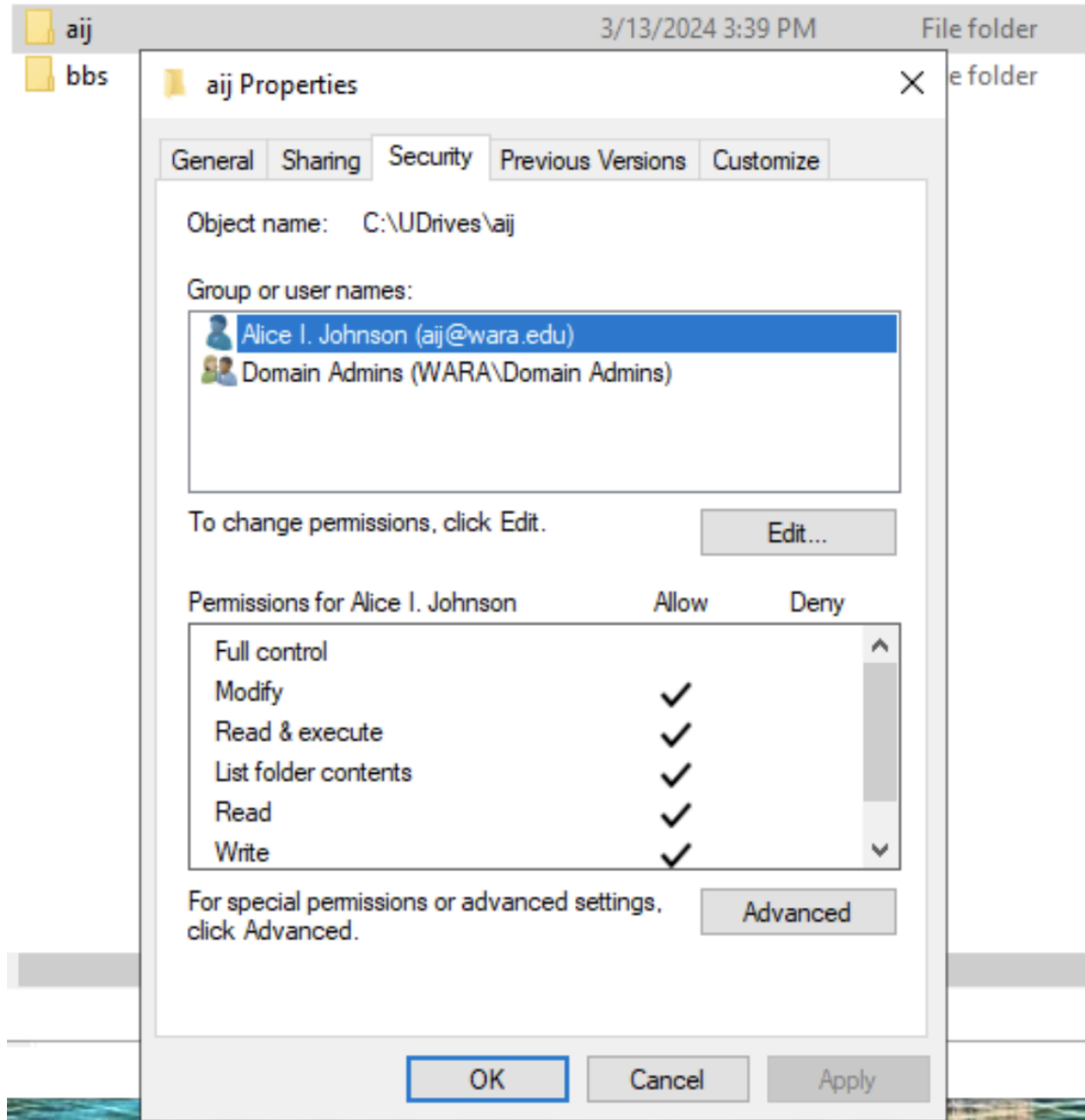


Figure 5: Group Policy updated and permissions configured for Faculty and Students groups.

Step 7: Verified Mapped Drive Access

Finally, I verified the mapped drive by opening **This PC** and confirming that the **S:** drive appeared.

I tested access using different users:

- A student user should have read-only access.
- A faculty user should have permission to modify the contents.
- Admin access was also tested to verify permissions.

The lab says to confirm that the S: drive is visible, test read-only access for Students, and test modify access for Faculty users.

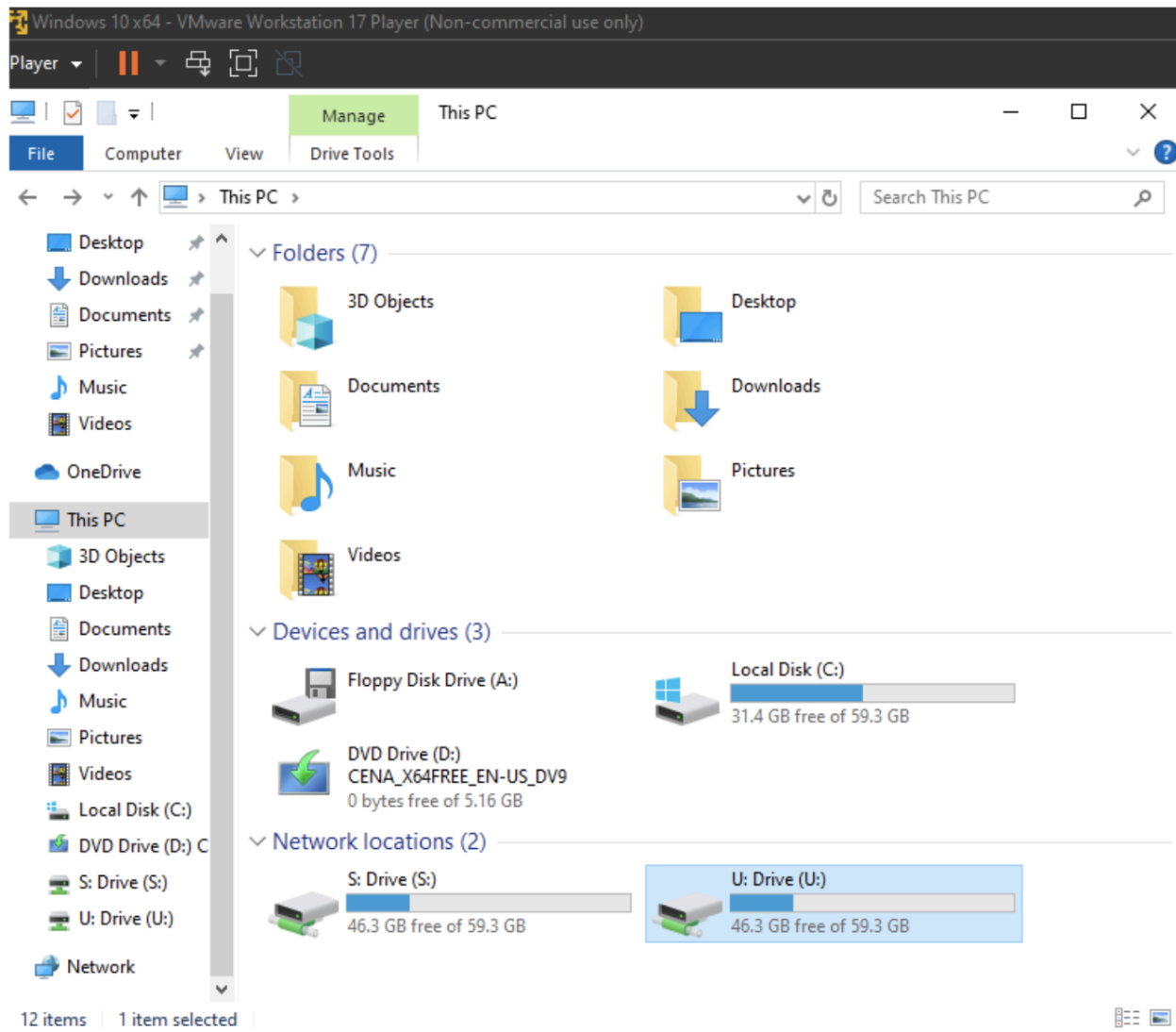


Figure 6: S: drive mapped successfully to the SecurityPlus shared folder.

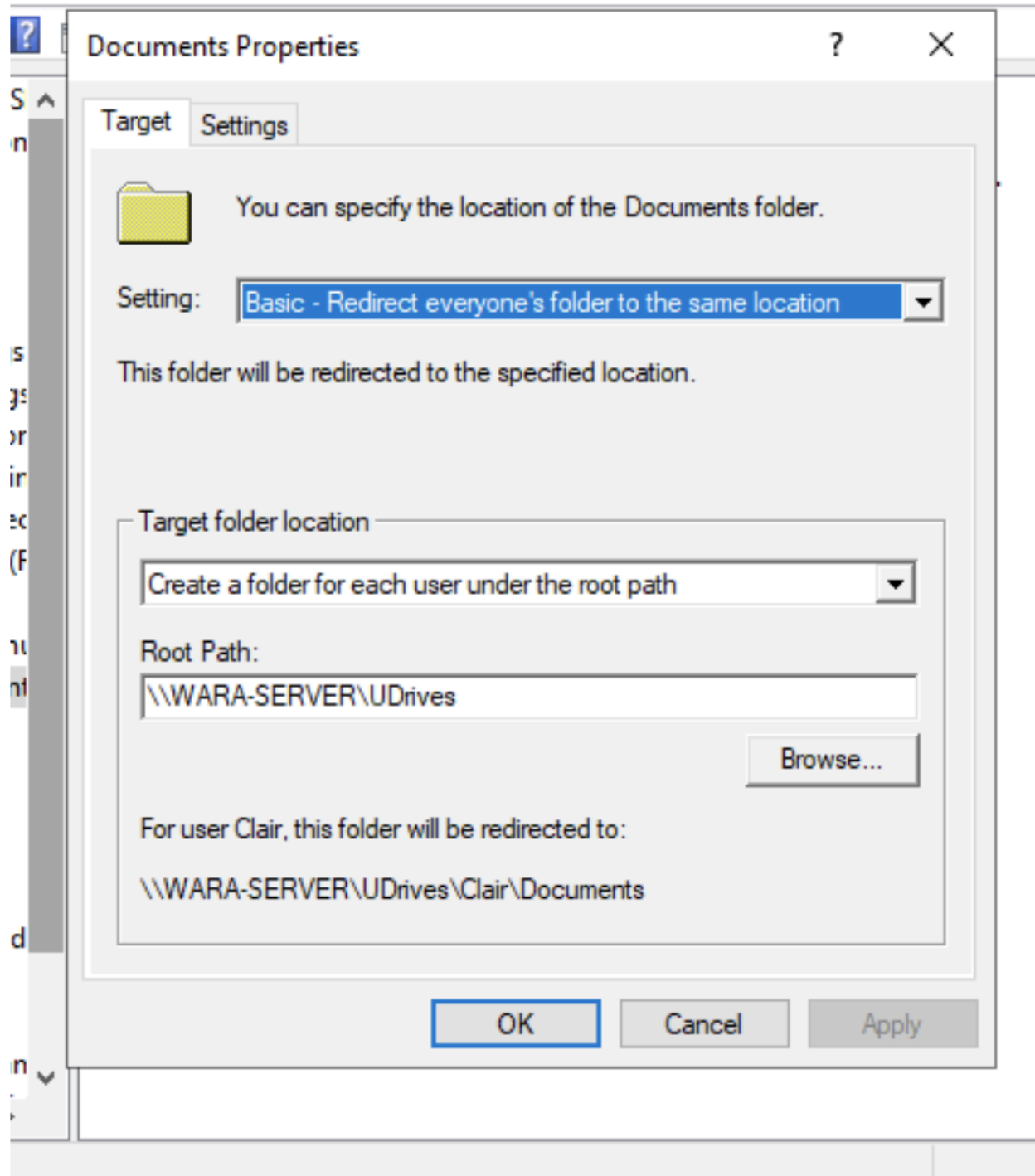


Figure 7: Documents folder redirection configured to store user files on the server.

7. Key Findings

Finding	Explanation
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Group Policy centralizes configuration	Desktop settings and mapped drives can be managed from the server.
User Configuration applies to users	Wallpaper, clean desktop, and mapped drive settings were configured under user policies.
UNC paths are important	Shared resources must use correct UNC paths like <code>\\WARA-SERVER\SecurityPlus</code> .
<code>gpupdate</code> applies policy changes	Running <code>gpupdate</code> helps apply new or updated GPO settings.
Mapped drives simplify access	Users can access shared folders through a drive letter instead of a network path.
Group permissions control access	Faculty and Students had different permission levels based on group membership.

8. Analysis

This lab showed how Group Policy can simplify IT administration in a domain environment. Instead of configuring each user's computer manually, an administrator can create a GPO and apply settings centrally.

The mapped drive portion of the lab is especially useful for IT support because many organizations use shared drives for department files. Mapping the `SecurityPlus` share as the `S:` drive makes it easier for users to find the shared resource.

This lab also showed that Group Policy and permissions must work together. Even if a drive is mapped successfully, users still need the correct share and NTFS permissions to access or modify files.

9. IT Support Relevance

This lab is highly relevant to IT Support Analyst, Help Desk Analyst, Desktop Support, and Junior System Administrator roles.

It connects to real job tasks such as:

- Troubleshooting Group Policy issues
- Running `gpupdate`
- Verifying whether a GPO applied

- Mapping network drives
- Troubleshooting missing network drives
- Checking user group membership
- Checking share and NTFS permissions
- Supporting department shared drives
- Managing desktop configuration policies
- Documenting policy changes

10. Skills Demonstrated

This lab demonstrates:

- Group Policy Management Console
- Group Policy Management Editor
- Creating and linking GPOs
- User Configuration policies
- Group Policy Preferences
- Drive Maps
- Mapped network drives
- UNC path configuration
- `gpupdate`
- Share permissions
- NTFS permissions
- Domain group access control
- Windows 10 client testing
- IT support documentation

11. Ticket-Style Summary

Issue:

Domain users needed centralized desktop configuration and easier access to the `SecurityPlus` shared folder.

Impact:

Without Group Policy, desktop settings and shared drive access would need to be configured manually on each user computer.

Steps Taken:

- Opened Group Policy Management Console.
- Created and linked a Desktop Wallpaper GPO.
- Configured wallpaper using the shared UNC path.
- Applied policy changes using `gpupdate`.

- Created and configured a Clean Desktop GPO.
- Linked the Clean Desktop GPO to the Students OU.
- Created an S: Drive GPO.
- Configured Drive Maps under Group Policy Preferences.
- Mapped `\\WARA-SERVER\SecurityPlus` to the `S:` drive.
- Created Faculty and Students groups.
- Assigned different share and NTFS permissions.
- Tested mapped drive visibility and access from domain users.

Root Cause:

Users needed centralized configuration and consistent access to shared resources.

Resolution:

Configured Group Policy Objects to apply desktop settings and map the `SecurityPlus` shared folder as the `S:` drive.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learned that Group Policy is a powerful tool for managing users and computers in a Windows domain environment. It allows administrators to apply settings like desktop wallpaper, desktop restrictions, and mapped drives without manually configuring every machine.

I also learned that mapped drives depend on both Group Policy and correct permissions. A drive may appear for the user, but access still depends on group membership, share permissions, and NTFS permissions.